

$$\text{NL}_{\mathbf{k}_s \rightarrow 0} \left(\begin{array}{c} \text{Diagram 1} \end{array} \right) = \text{Diagram 2} \times \text{Diagram 3} \times \text{Diagram 4}$$

The diagram on the left, enclosed in large parentheses, represents a four-point interaction. It features four external legs, each starting from a small square vertex and ending at a circular vertex. The external momenta are labeled $\delta h_{\mathbf{k}_1}$, $\delta h_{\mathbf{k}_2}$, $\delta h_{\mathbf{k}_3}$, and $\delta h_{\mathbf{k}_4}$. The internal structure consists of four wavy lines forming a square loop. The wavy lines are labeled with momentum and index: the top horizontal line is $A_{\mathbf{q}}^\mu$, the bottom horizontal line is $A_{\mathbf{q}+\mathbf{k}_s}^\mu$, the left vertical line is $A_{\mathbf{q}+\mathbf{k}_1}^\mu$, and the right vertical line is $A_{\mathbf{q}-\mathbf{k}_4}^\mu$.

This diagram is equal to the product of three diagrams:

- Diagram 2:** A vertical wavy line with two circular vertices at its ends. Each circular vertex is connected to a small square vertex. The top circular vertex also has a cross symbol ($\otimes$) on it.
- Diagram 3:** A closed loop of wavy lines. Two vertices on the loop are marked with solid black dots.
- Diagram 4:** A vertical wavy line with two circular vertices at its ends. Each circular vertex is connected to a small square vertex. Both circular vertices have a cross symbol ($\otimes$) on them.